

FUNCTION MaxProductOfThree(A)

∞^- = SET max1

∞^- = SET max2

∞^- = SET max3

∞^+ = SET min1

∞^+ = SET min2

FOR each number x in A DO

IF $x > \text{max1}$ THEN

max3 = max2

max2 = max1

max1 = x

ELSE IF $x > \text{max2}$ THEN

max3 = max2

max2 = x

ELSE IF $x > \text{max3}$ THEN

$\text{max3} = x$

END IF

IF $x < \text{min1}$ THEN

$\text{min2} = \text{min1}$

$\text{min1} = x$

ELSE IF $x < \text{min2}$ THEN

$\text{min2} = x$

END IF

END FOR

SET $\text{option1} = \text{max1} * \text{max2} * \text{max3}$

SET $\text{option2} = \text{max1} * \text{min1} * \text{min2}$

RETURN MAX(option1 , option2)

END FUNCTION